

Representation of Finite and Compact Groups

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Abstract
Representation theory is a field of Mathematics that deals with representing abstract algebraic structures as linear transformations of vector spaces. Essentially, it is helpful as it reduces the problems of Abstract Algebra down to Linear Algebra, which is a well understood field. A lot of applications of Representation Theory are often seen in other disciplines like Physics and Chemistry. In this write-up, we will give a brief introduction to results of representations of finite groups and then move on to look at representations of compact groups where we will look at the Peter Weyl theorem and prove one of its versions.
Preliminaries

Definition A representation of a group G is a continuous homomorphism $\rho : G \rightarrow GL_n(\mathbf{C})$. Given a group G of cardinality n , we can obtain the regular representation of degree n after action of G on itself by translation.

Definition A subrepresentation of a representation $\rho: G \rightarrow GL(V)$ is a vector subspace $W \subset V$ such that $\rho(g)(x) \in W$ for all $g \in G$ and $x \in W$.

Definition If a representation has no subrepresentations except for the trivial ones, then it is said to be irreducible.

Definition The character of a representation ρ is a function $\chi_\rho: G \rightarrow \mathbf{C}$ such that $g \mapsto \text{Tr}(\rho(g))$

Definition Let $\zeta, \epsilon \in \mathbf{C}^G$. Then we define the inner product of both functions as

$$\langle \zeta | \epsilon \rangle = \frac{1}{|G|} \sum_g \zeta \bar{\epsilon}$$

Definition Let X be a locally compact space and $\mathcal{B}$ be the σ algebra generated by the open subsets of X . A positive measure μ is said to be regular Borel if:

- $\mu(K) < \infty$ for every compact set K
- If $B \in \mathcal{B}$, then $\mu(B) = \inf\{\mu(O) : O \text{ open}, B \subseteq O\}$
- If $B \in \mathcal{B}$ and $\mu(B) < \infty$, then $\mu(B) = \sup\{\mu(K) : K \text{ compact}, K \subseteq B\}$

Definition An operator T on a Hilbert space H is said to be a compact operator if it carries the unit sphere (or any bounded subset of H) into a relatively compact set.

Spectral Theorem Let T be a compact and self adjoint operator on a complex Hilbert space H . Then there exists at most countable sequence $\lambda_1, \lambda_2, \dots$ of non zero reals that converge to 0 and an orthogonal decomposition

$$H = V_0 \oplus \bigoplus_n V_{\lambda_n}$$

where V_0 is the kernel of T and V_{λ_n} are finite dimensional eigenspaces of λ_n .

Important Results in Finite Groups
Maschke's Theorem Let $\rho: G \rightarrow GL(V)$ be a representation of G and let $W \subseteq V$ be a subrepresentation. Then there exists a complementary subrepresentation $U \subseteq V$ such that $U \oplus W = V$. Schur's Lemma Let $\rho_V: G \rightarrow GL(V)$ and $\rho_W: G \rightarrow GL(W)$ be irreducible representations of G. - Let $f: V \rightarrow W$ be a G linear map. Then either f is an isomorphism or f is the zero map. - Let $f: V \rightarrow V$ be a G linear map. Then $f = \lambda 1_V$ for some $\lambda \in \mathbf{C}$. Theorem Given any regular representation of a group G, we can decompose it as a direct sum of irreducible representations with each irreducible representation having as many direct sum copies of itself as its dimension i.e. $V_{reg} = U_1^{\oplus d_1} \oplus \dots \oplus U_r^{\oplus d_r}$ where U_i are the irreducible representations and d_i are their dimensions. Result Let χ be a character of a group G. Then χ is irreducible if and only if its inner product with itself is 1.

The Haar Measure
A left Haar measure on a locally compact group G is a positive regular Borel measure m such that $m(xE) = m(E)$ for every element x belonging to the group and set E belonging to the Borel sigma algebra $\mathcal{B}$. A right Haar measure is defined similarly. Any group in which the left and right Haar measures are equal is said to be unimodular.
Examples: - Suppose G is a finite group. Then for any subset E of G, we define $m(E) := \frac{1}{ G } E$. This is also known as the normalized Haar measure because the measure of the entire group is equal to 1. - If G is an infinite discrete group, then if E is a finite subset of G, we define $m(E) = E$ and if E is an infinite set, then $m(E) = \infty$. - If m_1 and m_2 are Haar measures defined on two groups G_1 and G_2, then $m_1 \times m_2$ is a Haar measure on $G_1 \times G_2$.
Lemma Let G be a locally compact group and let m be a regular Borel measure which is finite on all compact subsets of G . Then m is a left Haar measure if and only if $\mu(A) = m(A^{-1})$ is a right Haar measure.
Theorem Let G be a locally compact group. Then there exists a left Haar measure which is unique upto scalar multiplication.
Lemma Let m be a left invariant Haar measure on G . Then there exists a real valued continuous function ρ on G such that $m(Ex) = \rho(x)m(E)$ where $x \in G$ and E is Borel.
Theorem Any group that is compact or abelian is unimodular.

Convolution
Definition Let G be a locally compact group and let m denote the Haar measure on G . The convolution $f * g$ is said to exist at a point $x \in G$ if $f(y)g(y^{-1}x)$ is y -integrable and
$f * g(x) = \int f(y)g(y^{-1}x) dy$
exists almost everywhere on G .
Theorem Let G be unimodular group and let $f \in L^p(G), g \in L^q(G)$ where $1 \leq p \leq \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$. Then $f * g$ exists and is continuous with $\ f * g\ \leq \ f\ _p \ g\ _q$.
Compact Groups

Theorem Every finite dimensional representation of a compact group G by linear transformations is equivalent to a unitary representation.

Definition If ϕ is a finite dimensional representation of a group G by linear transformations on a vector space V , we define a family of functions called the representation functions coming from ϕ . They are functions F of the form $F(x) = f(\phi(x)v)$ where $x \in G, v \in V, f \in V^*$.

Definition Given a finite dimensional unitary representation ρ of a group G on a vector space V (considered as a Hilbert space by choosing any inner product) over $\mathbf{C}$, a matrix coefficient is any function $f_{u,v}: G \rightarrow \mathbf{C}$ that is of the form $f_{u,v}(g) = \langle \rho(g)u, v \rangle = \eta(\rho(g)v)$ where η is a continuous linear functional on V and $u, v \in V$.

Schur's Orthogonality Relations If ϕ and ψ are finite dimensional, inequivalent, irreducible and unitary representations of a compact group G , then all representation functions of ϕ are orthogonal to all representation functions of ψ .

Definition Let $\epsilon > 0$ and let $f \in L^2(G)$. An element $\delta \in L^2(G)$ is said to be an approximate identity for f with respect to ϵ if and only if $\|f * \delta - f\| < \epsilon$ and $\|\delta * f - f\| < \epsilon$.

Definition The right regular representation is the representation R with the representation space $L^2(G)$ and $R(x)(f) = f^x, x \in G$ i.e $(R(x)f)(y) = f^x(y) = f(yx)$.

Definition Let G be a compact group. Then for each $f \in L^2(G)$, define $\mathcal{L}_f(g) = f * g$ where $g \in L^2(G)$.

Theorem $\mathcal{L}_f$ is bounded and compact for each $f \in L^2(G)$

The Peter Weyl Theorem
If G is a compact, topological group, R its right regular representation, f any self adjoint element of $L^2(G)$, (λ_i) the non zero eigenvalues of $\mathcal{L}_f$ and H_i the corresponding eigenspaces, then: - Each H_i is a finite dimensional invariant subspace of R and is also invariant under the right multiplication $*$ by any g in $L^2(G)$. - If R^i is the representation of G induced on H_i by R, then each eigenfunction of the eigenvalue λ_i is a representation function of R^i. f is an L^2 limit of finite linear combination of representation functions from the R^i. - If $f \in C_C(G)$, then f is the uniform limit of finite linear combinations of representation functions from the R^i.

Other Statements
Definition Given a set X , a collection of complex functions $\mathcal{A}$ is said to be an algebra if it is closed under addition, function multiplication and scalar multiplication. $\mathcal{A}$ is said to be unital if it has an identity element with respect to function multiplication.
Definition Let $\mathcal{A}$ be a collection of functions from a set X to Y . $\mathcal{A}$ separates points on X if for all distinct points x_1 and x_2 in X , there exists a function f in $\mathcal{A}$ such that $f(x_1) \neq f(x_2)$.
Stone Weierstrass Theorem Suppose X is a compact Hausdorff space and S is a separating subset of $C(X, \mathbf{C})$. Then the complex unital algebra generated by S is dense in $C(X, \mathbf{C})$
The next theorems are often considered as other statements of Peter Weyl's Theorem .
Theorem Let G be a compact Hausdorff group. Then the space of matrix coefficients is dense in $C(G)$ where $C(G)$ denotes the space of all continuous, complex valued functions on G .
Proof The proof of this theorem revolves around introducing an operator defined for every $y \in G$:

$$\begin{aligned} \tau(y) : L^2(G) &\rightarrow L^2(G) \\ \tau(y)f(x) &:= f(y^{-1}x) \end{aligned}$$

and noticing that for any non identity element $g \in G$, there exists a finite dimensional invariant subspace of $L^2(G)$ on which $\tau(g)$ is not the identity. This helps us in choosing a function $\phi \in H_\lambda$ where H_λ is a non zero eigenspace of $\mathcal{L}_f$ such that

$$\tau(g)\phi \neq \phi$$

which consequently helps us to define a matrix coefficient that separates points in G . Then by an application of Stone Weierstrass theorem, we are able to get the conclusion of the theorem.

Theorem $L^2(G)$ decomposes into an orthogonal direct sum of all the irreducible unitary representations in which the multiplicity of each irreducible representation is equal to its degree.

References
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